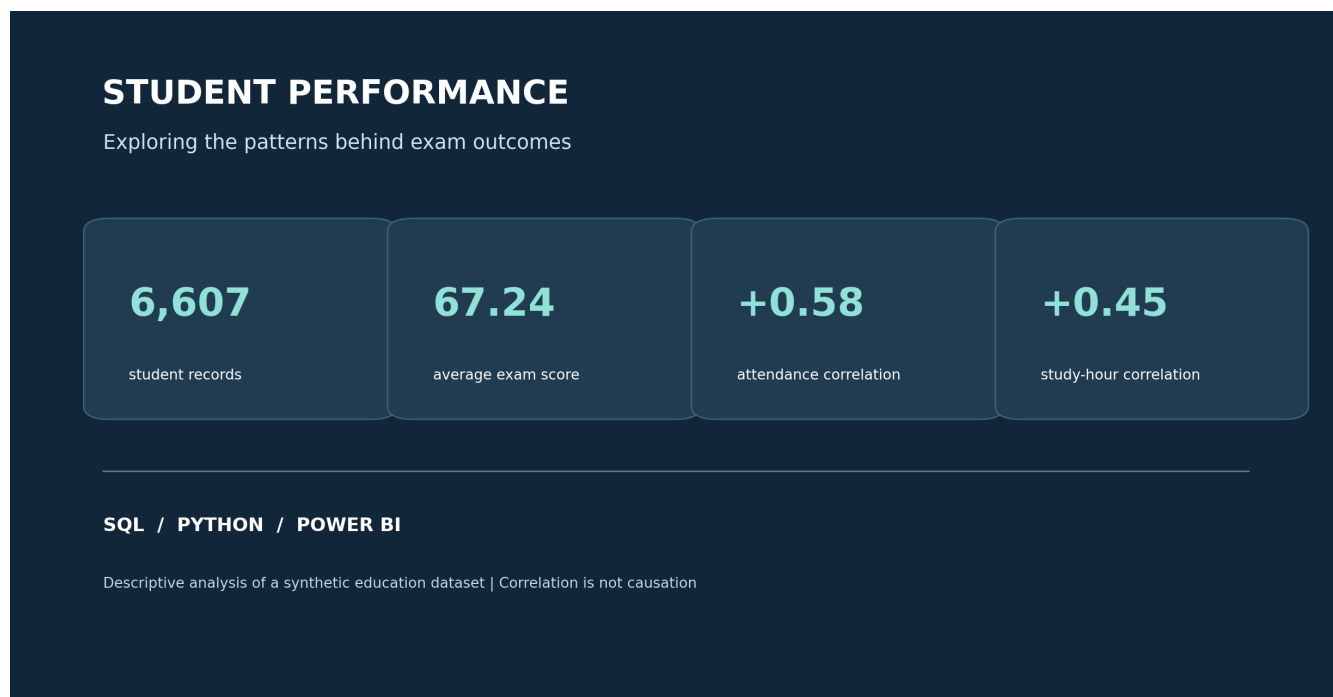


What patterns shape exam scores?

A descriptive portfolio case study of **6,607 synthetic student records** using PostgreSQL, Python and a Power BI-ready model. It examines study habits, attendance, tutoring and sleep without making causal claims.



Three observations from these records

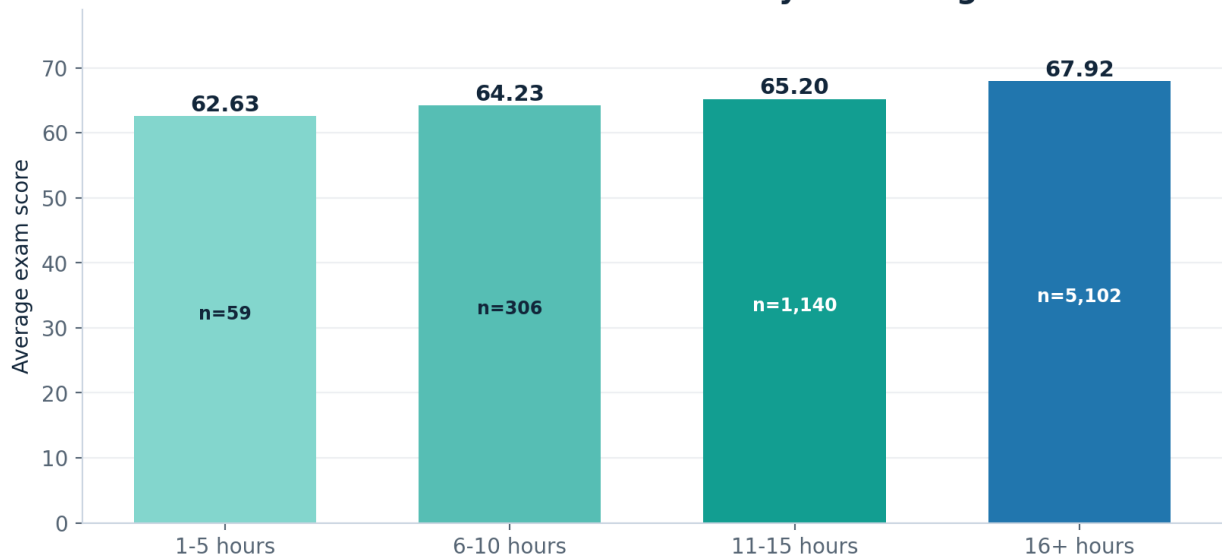
- **Attendance:** the mean exam score rises from 64.21 (60-69% attendance) to 70.14 (90-100% attendance).
- **Study time:** students in the 16+ weekly-hour band average 67.92; those in the 1-5-hour band average 62.63, although the latter contains only 59 records.
- **Sleep:** the observed linear correlation with score is approximately -0.02; this synthetic sample offers little evidence of an unadjusted linear pattern.

Interpretation: relationships do not establish cause and effect. The source is synthetic, not a representative school sample.

Study habits and attendance

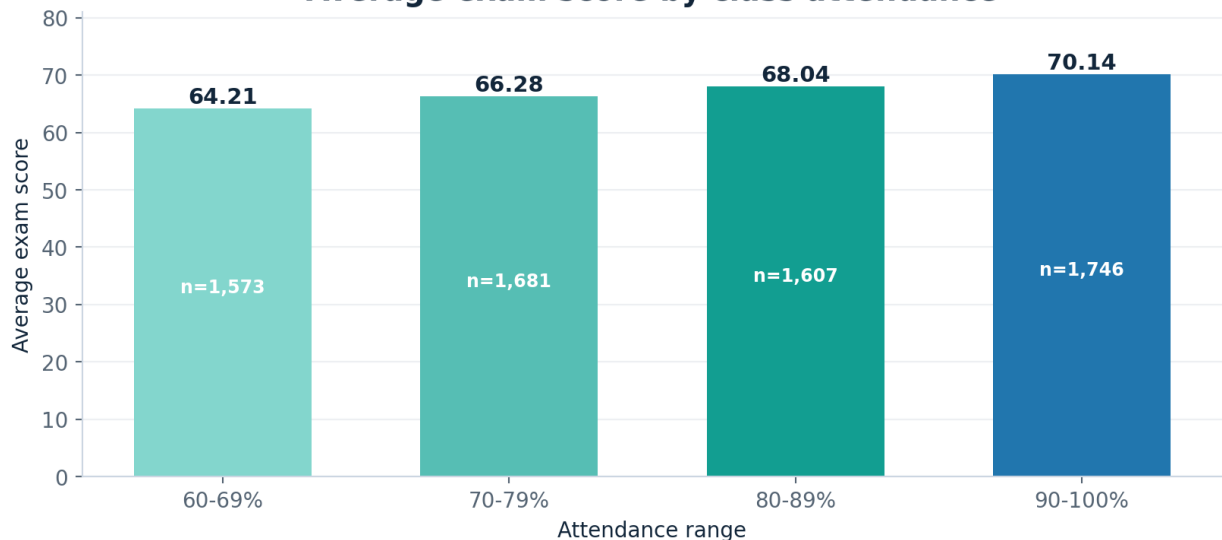
The broad study-hour and attendance bands display gradual average-score differences. Each chart labels group sizes, because a precise average based on 59 observations is less stable than one based on thousands.

Exam score rises across study-hour ranges



Study-hour gap: $67.92 - 62.63 =$ approximately **5.30 points** between extreme groups; this unadjusted contrast is not a recommended study-hour target.

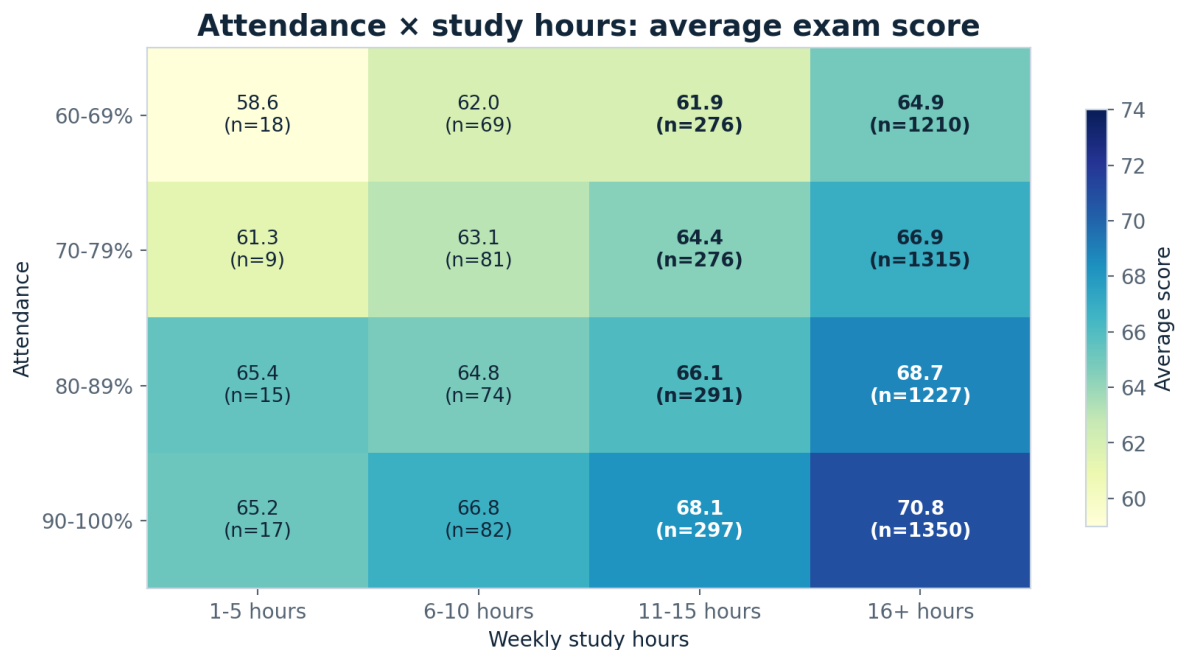
Average exam score by class attendance



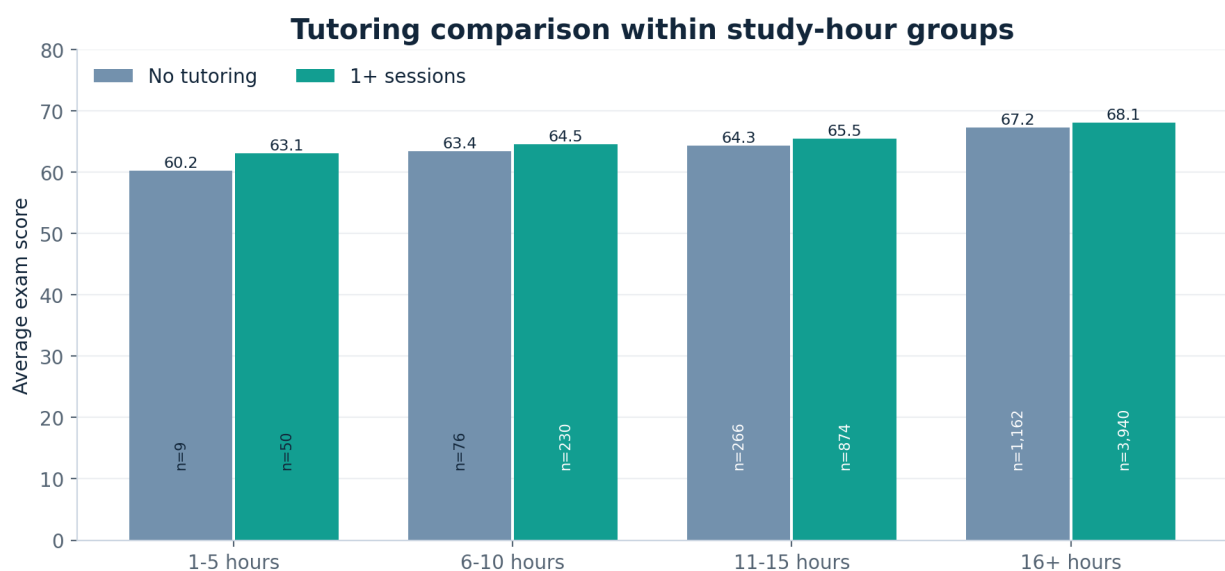
Attendance gap: the 90-100% band averages approximately **5.93 points** above the 60-69% band. Other factors may differ across groups.

Explore combined patterns

A more useful analytical question is whether the observed patterns persist when comparing more similar study-hour groups. These charts expose the joint attendance-by-study matrix and tutoring differences within study-hour strata.



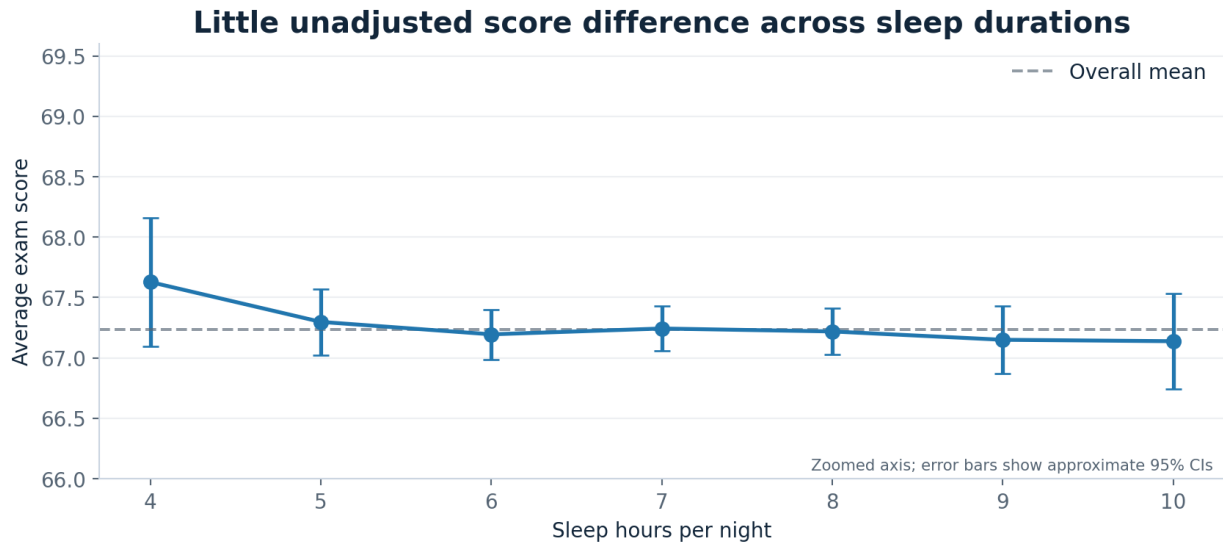
Read the sample sizes: counts are printed in every cell; compare within rows and columns rather than claiming a particular habit compensates for another.



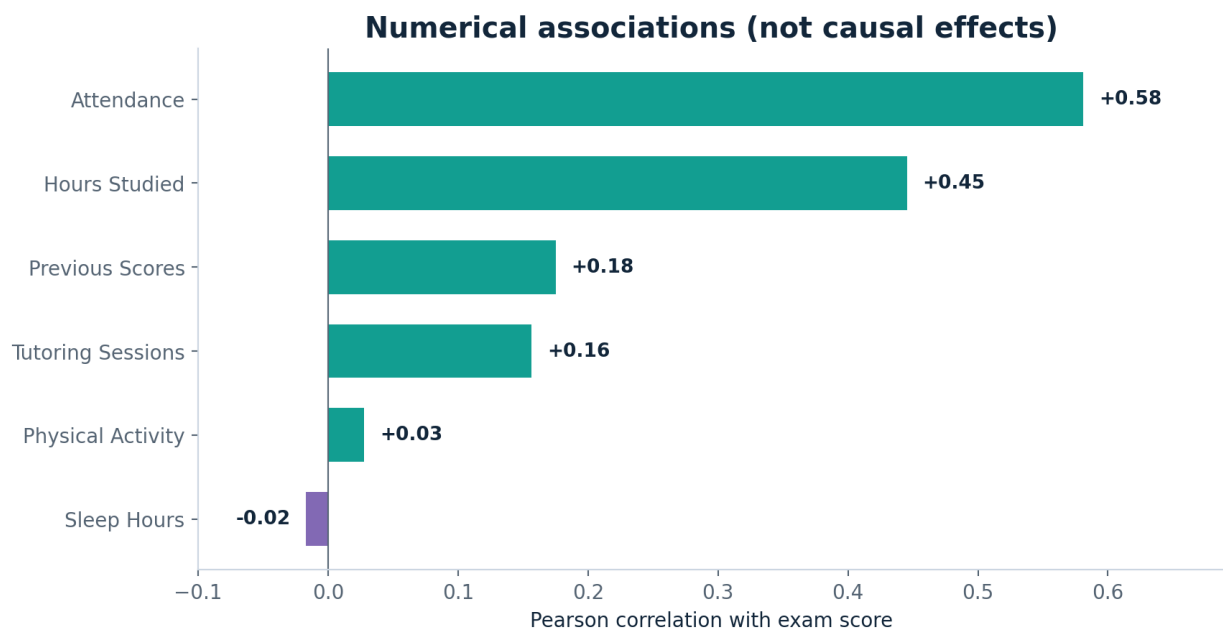
Tutoring comparison: study-hour stratification helps contextualize the raw difference, but does not control for attendance, prior scores, resources or selection into tutoring.

Wellbeing and correlations

Near-zero correlation is a finding in this dataset, not proof that sleep is irrelevant to student health or educational outcomes. The sleep chart deliberately indicates its zoomed axis and uncertainty bars.



The unadjusted sleep correlation is **-0.017**; small observed mean differences are close to the noise level. Approximate intervals are shown to discourage overinterpretation.



Pearson correlation measures one type of association. It does not describe an independent causal contribution or establish how a variable would behave in real schools.

Method and reproducibility

Workflow

1. Ingest the original 6,607-row, 20-column CSV without changing source exam scores.
2. Execute three original PostgreSQL assignment queries and five optional investigation/validation queries.
3. Prepare grouped summaries and six labeled charts with pandas and matplotlib.
4. Export a 27-column Power BI-ready table with explicit categorical Unknown values, grouping fields and a score quality flag.
5. Create a Power BI report with the supplied DAX measures, theme, page plan and QA checklist.

Source data quality

- 6,607 rows, 20 raw variables and no exact duplicate rows.
- Three original partially missing fields: teacher quality (78), parental education (90), distance from home (67).
- One score = 101 despite the conventional 0-100 scale: original assignment outputs retain it; the dashboard can filter or flag it.
- The overall average is 67.2357 with this flagged record and 67.2305 when it is excluded.

Optional predictive illustration

An 80/20 random held-out split of synthetic rows gives test MAE approximately **1.27** for a numerical ridge baseline and **1.23** after adding two categorical fields (teacher quality and extracurricular participation). This is a demonstration of evaluation mechanics, not a deployable assessment model.

Important boundaries

This dataset is described as **synthetic**, and no experimental treatment is assigned. Missing categories are preserved as Unknown. Neither predictive fit nor correlation is evidence of intervention effectiveness; source data are not representative of real students.

How to review or rebuild

The repository contains an executed Jupyter notebook, exact SQL and CSV query outputs, original source, Python scripts, an offline HTML report, and a Power BI build guide. The final native **.pbix must be saved from Power BI Desktop** using the prepared assets.